

Marcel Walter

POSTDOCTORAL RESEARCHER · SENIOR QUANTUM SOFTWARE ENGINEER

Munich, Germany

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Education

Technical University of Munich (TUM)

HABILITATION CANDIDATE

Munich, Germany

2027 (expected graduation)

- Thesis: *Advanced Design Automation and Simulation for Atomic-Scale Computing*

University of Bremen

PH.D. IN COMPUTER SCIENCE (SUMMA CUM LAUDE)

Bremen, Germany

2016 – 2021

- Thesis: *Design Automation for Field-coupled Nanotechnologies*

University of Bremen

M.SC. IN COMPUTER SCIENCE

Bremen, Germany

2015 – 2016

- Specialization in *Safety, Security, and Quality*
- Thesis: *Exact Synthesis of Abstraction-Level-Independent Logic Networks*

University of Bremen

B.SC. IN COMPUTER SCIENCE

Bremen, Germany

2011 – 2015

- Thesis: *Optimization of Quantum Circuits for Linear Nearest Neighbor Architectures*

Experience

Munich Quantum Software Company (MQSC)

SENIOR QUANTUM SOFTWARE ENGINEER

Munich, Germany

2025 – Present

- Industrial-strength software for quantum computers
- HPCQC integration

University of Bremen

SUBSTITUTE PROFESSOR

Bremen, Germany

2024

- Teaching graduate courses concerning computational nanotechnology, logic synthesis, and physical design

Technical University of Munich (TUM)

POSTDOCTORAL RESEARCHER

Munich, Germany

2021 – Present

- Design Automation for post-CMOS Emerging Nanotechnologies
- Logic Synthesis & Physical Design
- Silicon Photonics
- Superconducting Quantum Computing
- Advising theses of all levels
- Teaching undergraduate and graduate courses
- Supervisor: Prof. Dr. Robert Wille, Chair for Design Automation

University of Bremen

RESEARCHER AND PH.D. STUDENT

Bremen, Germany

2016 – 2021

- Physical design of post-CMOS Emerging Nanotechnologies
- Verification of CRAM plans for PR2 robots at the collaborative research center EASE
- Teaching undergraduate courses and supervising Bachelor's and Master's theses
- Supervisor: Prof. Dr. Rolf Drechsler, Research Group of Computer Architecture

University of Bremen

INDEPENDENT STUDY

Bremen, Germany

2016

- Reactive system synthesis from LTL specifications using Satisfiability Modulo Theories (SMT)
- Supervisor: Prof. Dr. Rüdiger Ehlers, Modelling of Technical Systems Research Group

German Research Center for Artificial Intelligence (DFKI)

RESEARCH ASSISTANT

Bremen, Germany

2015 – 2016

- Formal verification of access control systems
- Supervisor: Prof. Dr. Christoph Lüth, Cyber Physical Systems

- Practical Computer Science 1 (Java basics, OOP, algorithms and data structures)
- *Supervisor*: Dr. Karsten Hölscher, Software Engineering Group
- Technical Computer Science 1 (Computer architecture, Boolean Algebra, logic synthesis)
- *Supervisor*: Prof. Dr. Rolf Drechsler, Research Group of Computer Architecture

Research Interests

- **Design Automation for Post-CMOS Nanotechnologies**: Field-coupled Nanocomputing (Quantum-dot Cellular Automata, Silicon Dangling Bonds) — logic synthesis, physical design, placement & routing, verification, and physical simulation.
- **Quantum Computing**: software and hardware integration for quantum computers, including cryogenic CMOS control electronics and HPC-quantum computing (HPCQC) integration.
- **Emerging Substrates**: silicon photonics and superconducting quantum circuit design, extending the same design-automation methodology to new physical substrates.
- **Algorithmic Foundations**: exact and heuristic combinatorial optimization, SAT/SMT-based synthesis and verification, graph algorithms, and computational complexity theory.

Languages

German (Native) · English (Fluent) · Dutch (B2)

Honors & Awards

- 2025 **Best Student Paper Award**, IEEE NANO 2025
- 2025 **Best Paper Award Candidate**, LASCAS 2025
- 2024 **Teaching Award Candidate (*Studierendenpreis*)**, Berninghausenpreis 2024
- 2024 **Best Presentation Award**, IEEE NANO 2024
- 2023 **Best Paper Award Candidate**, IWLS 2023
- 2021 **Best Paper Award Candidate**, ASP-DAC 2021
- 2020 **Best Young Student Fellow Research Video Award**, DAC 2020
- 2020 **Young Student Fellow Program Scholarship**, DAC 2020
- 2020 **Best Student Forum Paper Award**, ISVLSI 2020
- 2020 **Best Paper Award Candidate**, ISVLSI 2020
- 2019 **Best Research Demo Award**, ISVLSI 2019

Talks

Ningbo University

Ningbo, China

INVITED TALK

Jul. 2026

- From Logic to Atoms: Design Automation for Atomic-Scale Computing

Free Silicon Conference

Frankfurt (Oder), Germany

INVITED TALK

Jul. 2025

- aigverse: Toward Machine Learning-Driven Logic Synthesis

4th Annual Hack4Her Event

Amsterdam, Netherlands

WORKSHOP WITH HANDS-ON

Jun. 2025

- Chip Design for AI

Universidade Federal de Minas Gerais - Workshop on Nanocomputing

Belo Horizonte, Brazil

INVITED WORKSHOP CONTRIBUTION

Feb. 2025

- The Munich Nanotech Toolkit (MNT)

IEEE International Conference on Nanotechnology (IEEE NANO) 2024

Gijón, Spain

SPECIAL SESSION HOST AND ORGANIZER

Jul. 2024

- Field-coupled Nanocomputing (FCN)

3rd Annual Hack4Her Event

Amsterdam, Netherlands

WORKSHOP WITH HANDS-ON

Jun. 2024

- Logic Synthesis for AI

International Workshop on Logic and Synthesis (IWLS) 2024

INVITED SPECIAL SESSION

- The Case for Planar Logic Synthesis – Crossing Costs in Nanotech

Zurich, Switzerland

Jun. 2024

University of Alberta

GUEST RESEARCHER LECTURE

- The Munich Nanotech Toolkit (MNT)

Edmonton, Canada

Mar. 2024

EPFL - Laboratoire des Systèmes Intégrés

THESIS OVERVIEW

- Design Automation for Field-coupled Nanotechnologies

Lausanne, Switzerland

May 2021

Johannes Kepler University (JKU) Linz

GUEST LECTURE

- Field-coupled Nanocomputing (FCN)

Linz, Austria

Jan. 2019

Teaching Experience

University of Bremen

GRADUATE COURSES

- Logic Synthesis & Physical Design
- Unconventional Computing: Nanotech and Biochips
- Seminar: Computational Nanotechnologies

Bremen, Germany

2024

Technical University of Munich (TUM)

UNDERGRADUATE & GRADUATE COURSES

- Introduction to Emerging Computing Technologies
- Logic Synthesis & Physical Design

Munich, Germany

2022 – Present

University of Bremen

ADVANCED TOPICS IN TECHNICAL COMPUTER SCIENCE (SEMINAR)

- Co-led with Prof. Dr. Rolf Drechsler; Boolean Algebra, Logic Synthesis, Graph Theory, hardware description languages, and post-CMOS emerging technologies

Bremen, Germany

2017 – 2020

University of Bremen

HEAD OF ORGANIZATION, TECHNICAL COMPUTER SCIENCE 1

- Lecture hall management, briefing tutors, addressing student concerns, creating and correcting assignment sheets, proxy lectures, and organizing exams

Bremen, Germany

2017 – 2018

University of Bremen

UNDERGRADUATE COURSES

- Technical Computer Science 1 (Computer Architecture, Logic Synthesis)
- Practical Computer Science 1 (Java, Algorithms and Data Structures)

Bremen, Germany

2013 – 2021

Supervision & Mentoring

PH.D. CANDIDATES

Technical University of Munich (TUM)

JAN DREWNIOK — PH.D. (*summa cum laude*)

- Thesis: Design and Simulation of Atomic-Scale Computing: A Comprehensive Framework for Physical Simulation, Logic Design, Logic Analysis, and Circuit Design
- IEEE Nanotechnology Council (NTC) 2026 Best PhD Thesis Award
- DATE 2025 Best Poster Award at the PhD Forum
- IEEE NANO 2024 Student Design Competition Winner: *NanoLogic: An Educational App for Atomic-Scale Computing*

2026

Munich, Germany

SIMON HOFMANN — PH.D. (*summa cum laude*)

- Thesis: Physical Design for Field-coupled Nanocomputing
- LASCAS 2025 Best Paper Award Candidate: *Physical Design for Field-coupled Nanocomputing with Discretionary Cost Objectives*

2026

BACHELOR'S & MASTER'S THESES (SELECTION)

2025	Michael Feldmeier (M.Sc.) , Exploring Feedback-Guided AIG Optimization to Align Logic Synthesis with Mapping Objectives	TUM
2025	Danilo Quinci (M.Sc.) , Design of Standard Cell Libraries for Molecular Field-coupled Nanocomputing Design Automation	PoliTo
2025	Felix Kiefhaber (B.Sc.) , Super-Tile Routing for Omnidirectional Information Flow in Silicon Dangling Bond Logic	TUM
2024	Willem Lambooy (M.Sc.) , Hierarchically Bounded Composable Interactions: Applications and Extensions to Atom-Scale Logic Design	Radboud
2023	Benjamin Hien (M.Sc.) , Signal Distribution Networks in Automatic QCA Standard-Cell Placement & Routing	TUM
2023	Isabella Venancia Gardner (B.Sc.) , Machine Learning for Electrostatic Ground State Simulation of Silicon Dangling Bond Logic	VU Amsterdam
2021	Sophia Kuhn (B.Sc.) , Design Understanding für Quantum-dot Cellular Automata Layouts	Uni Bremen
2021	Daniel Staack (B.Sc.) , Fehlerfindung in robotischen Plänen mittels Fuzzing	Uni Bremen
2020	Till Schlechtweg (B.Sc.) , Gate-level Placement für Field-coupled Nanocomputing unter Berücksichtigung von Clocking Constraints	Uni Bremen

Open-Source Projects (Selection)

Munich Nanotech Toolkit (MNT)

<https://github.com/cda-tum/fiction>

INITIATOR AND CORE MAINTAINER

2018 – Present

- Initiated and maintain the Munich Nanotech Toolkit, an ecosystem of open-source tools for design automation of post-CMOS nanotechnologies (cf. the tool paper [46]), contributing across its components as needed.
- Flagship project: fiction, a framework for Design Automation of Field-coupled Nanotechnologies for researchers and developers (cf. the tool paper [83]), honored with the ISVLSI 2019 Best Research Demo Award.

aigverse

<https://github.com/marcelwa/aigverse>

INITIATOR AND MAINTAINER

2024 – Present

- A Python library for working with logic networks, synthesis, and optimization.
- High-performance C++ backend and interoperability with other logic synthesis tools.
- Integration with popular data science and machine learning libraries.

Munich Quantum Toolkit (MQT)

<https://github.com/munich-quantum-toolkit/core>

CONTRIBUTOR

2025 – Present

- Contributing across repositories of the Munich Quantum Toolkit as part of ongoing quantum software engineering work at MQSC.
- Exemplary repository: MQT Core, the toolkit's foundational C++/Python library providing the shared data structures and algorithms (compilation, simulation, verification) that its other quantum-software components build on (cf. the tool paper).

EPFL Logic Synthesis Libraries

<https://github.com/lsils/mockturtle>

CONTRIBUTOR

2020 – Present

- Contributing to the EPFL Logic Synthesis Libraries, a collection of C++ libraries for logic network synthesis, optimization, and verification.
- Exemplary repository: mockturtle, a header-only library for logic network manipulation and optimization.

QDMI-on-IQM

<https://github.com/iqm-finland/QDMI-on-IQM>

CONTRIBUTOR AND CO-MAINTAINER

2025 – Present

- Implementation of the Quantum Device Management Interface (QDMI) for IQM's quantum computing hardware, the standard interface connecting HPC software stacks to quantum backends.
- Enables the HPCQC integration case study demonstrated on real IQM hardware [81].
- Co-maintain the repository: reviewing contributions and keeping the integration current with both QDMI and IQM's hardware interfaces.

EXECUTIVE TEAM AND ORGANIZING COMMITTEES

- 2027 **TPC Member**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2027 **Program co-Chair and TPC Member**, International Workshop on Logic and Synthesis (IWLS)
- 2026 **Best Poster Award Judge**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2026 **Publicity co-Chair, TPC Member, and Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2025 **Track Co-Chair**, IEEE CASS Latin American Symposium on Circuits and Systems (LASCAS)
- 2024 **Proceedings Chair, TPC Member, and Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2024 **Special Session Host and Organizer**, IEEE International Conference on Nanotechnology (IEEE NANO)
- 2023 **TPC Member**, International Conference on Computer-Aided Design (ICCAD)
- 2023 **Session Chair**, International Workshop on Logic and Synthesis (IWLS)
- 2020 **Technical Operator**, International Conference on Computer-Aided Design (ICCAD)
- 2018 **Staff Member**, European Test Symposium (ETS)

REVIEWER FOR INTERNATIONAL CONFERENCES AND JOURNALS

- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- Design Automation Conference (DAC)
- International Conference On Computer Aided Design (ICCAD)
- Design, Automation and Test in Europe (DATE)
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- ACM Journal on Emerging Technologies in Computing (JETC)
- IEEE Transactions on Emerging Topics in Computing (TETC)
- IEEE Transactions on Nanotechnology (TNANO)
- IEEE Transactions on Circuits and Systems (TCAS)
- IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
- Journal of Computational Electronics (JCEL)
- International Workshop on Logic and Synthesis (IWLS)
- ACM Great Lakes Symposium on VLSI (GLSVLSI)
- Euromicro Conference on Digital System Design (DSD)
- Microprocessors and Microsystems (MICPRO)
- Analog Integrated Circuits and Signal Processing (ALOG)
- International Conference on VLSI Design (VLSID)
- European Test Symposium (ETS)
- IEEE Design & Test (D&T)
- IEEE International Symposium on Multiple-Valued Logic (IS-MVL)
- IEEE International Symposium on Embedded Computing and System Design (ISED)
- IEEE CASS Latin American Symposium on Circuits and Systems (LASCAS)
- Mathematics
- EPJ Plus
- Integration - The VLSI Journal
- IEEE Embedded Systems Letters (ESL)

COMMUNITY SERVICE

2022–24 **Expert and Mentor**, Hack4Her Event / Technology Retreat / Hackathon

*Amsterdam,
Netherlands*

2013 –
Present **Student Mentor**, Gymnasium Syke

Syke, Germany

Challenges, Tutorials & Contests

ISPD Contest

PARTICIPANT AND MENTOR

2026

- Post-Placement Buffering and Sizing.

IWLS Programming Contests

PARTICIPANT AND MENTOR

2024

- Optimal synthesis of AIGs and XAIGs from truth table specifications.

High Level Synthesis Tutorial by Cadence @ DAC 2020

PARTICIPANT

2020

- Logic synthesis of a CNN AI Accelerator for Image Recognition.

Bremen Big Data Challenge

PARTICIPANT

2016

- Applied machine learning on League of Legends statistics to predict winning teams.

Books

- [1] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, *Design Automation for Field-coupled Nanotechnologies*. Springer, 2022. doi: 10.1007/978-3-030-89952-3

Journal Articles

- [2] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, “RTL-to-Atoms Synthesis of a Machine Learning Accelerator on Atomic-Scale Computers,” *IEEE Transactions on Nanotechnology (TNANO)*, 2026. doi: 10.1109/TNANO.2026.3689032
- [3] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “QuickCell: Fast Automatic Design of Standard Cells for Silicon Dangling Bond Logic,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. doi: 10.1109/TCAD.2025.3605537
- [4] S. Hofmann, **M. Walter**, R. Wille, “Efficient and Scalable Post-Layout Optimization for Field-coupled Nanotechnologies,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. doi: 10.1109/TCAD.2025.3549354
- [5] S. Hofmann, **M. Walter**, R. Wille, “Graph-Oriented Layout Design for Field-coupled Nanocomputing via Parallel Multi-Objective Search Space Exploration,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2025. doi: 10.1109/TCSI.2025.3614709
- [6] S. S. H. Ng, J. Croshaw, **M. Walter**, R. Wille, R. Wolkow, K. Walus, “Simulating Charged Defects in Silicon Dangling Bond Logic Systems to Evaluate Logic Robustness,” *IEEE Transactions on Nanotechnology (TNANO)*, 2024. doi: 10.1109/TNANO.2024.3372946
- [7] M. D. Vieira, S. S. H. Ng, **M. Walter**, R. Wille, K. Walus, R. S. Ferreira, O. P. Vilela Neto, J. A. M. Nacif, “Three-Input NPN Class Gate Library for Atomic Silicon Quantum Dots,” *IEEE Design & Test*, 2022. doi: 10.1109/MDAT.2022.3189814
- [8] F. Sill Torres, P. A. Silva, G. Fontes, **M. Walter**, J. A. M. Nacif, R. S. Ferreira, O. P. V. Neto, J. F. Chaves, R. Wille, P. Niemann, D. Große, R. Drechsler, “On the Impact of the Synchronization Constraint and Interconnections in Quantum-dot Cellular Automata,” *Microprocessors and Microsystems (MICPRO)*, 2020. doi: 10.1016/j.micpro.2020.103109
- [9] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, “Placement and Routing for Tile-based Field-coupled Nanocomputing Circuits is NP-complete (Research Note),” *Journal on Emerging Technologies in Computing Systems (JETC)*, 2019. doi: 10.1145/3312661

Conference Proceedings

- [10] J. Drewniok, **M. Walter**, R. Wille, “Operational Domain Sketching for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [11] M. Feldmeier, **M. Walter**, R. Wille, “A Framework for Post-Mapping Feedback-Guided Selective Logic Transformations,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [12] I. V. Gardner, **M. Walter**, R. Wille, M. Cochez, “AIG2PT: A Generative Pre-trained Transformer for Unconditional And-Inverter Graph Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [13] B. Hien, **M. Walter**, R. Wille, “Minimizing Area and Delay in Planar Physical Design for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [14] B. Hien, **M. Walter**, R. Wille, “Toward Compact Planar Circuits for Atomic-Scale Computing: A Hybrid Crossing Elimination Flow,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [15] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Holistic Physical Design for Silicon Atomic Logic: Cramming More Components onto Clock Zones,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [16] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Mastering the Exponential Complexity of Exact Physical Simulation of Silicon Dangling Bonds,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026.
- [17] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, “Design and Emulation Methodology for Atomic-Scale Systolic Arrays: An LLM Accelerator Case Study in Silicon DB Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [18] S. S. H. Ng, **M. Walter**, M. Yuan, J. Drewniok, R. H. Foote, R. Lupoiu, J. A. Fan, R. Wolkow, R. Wille, K. Walus, “Bit-PlanarNet: Learning-to-Layout Planar Gate Networks for Emerging Devices,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

- [19] **M. Walter**, M. Feldmeier, R. Wille, “Exact Synthesis with Optimal Switching Activity,” in *Design, Automation and Test in Europe (DATE)*, 2026.
- [20] **M. Walter**, I. V. Gardner, R. Wille, “aigverse: A Unified Infrastructure for Machine Learning-Driven Logic Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [21] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Towards Fast Automatic Design of Silicon Dangling Bond Logic,” in *Design, Automation and Test in Europe (DATE)*, 2025.
- [22] J. Drewniok, **M. Walter**, R. Wille, “QuickTrace: An Efficient Contour Tracing Algorithm for Defect Robustness Simulation of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.
- [23] I. V. Gardner, **M. Walter**, Y. Miyasaka, R. Wille, M. Cochez, “Bias by Design: Diversity Quantification to Mitigate Structural Bias Effects in AIG Logic Optimization,” in *Design, Automation and Test in Europe (DATE)*, 2025. doi: 10.23919/DATE64628.2025.10993009
- [24] B. Hien, D. Quinci, Y. Ardesi, G. Beretta, F. Ravera, **M. Walter**, R. Wille, “Bridging the Gap Between Molecular FCN and Design Automation with SIM(7)-MolPDK: A Physically Simulated Standard-Cell Library,” in *IEEE Latin American Conference on Nanotechnology (IEEE LANANO)*, 2025.
- [25] B. Hien, **M. Walter**, A. Tempia Calvino, A. Mishchenko, R. Wille, “Ashenhurst-Curtis Decomposition Using Don’t Cares,” in *International Workshop on Logic Synthesis (IWLS)*, 2025.
- [26] B. Hien, **M. Walter**, S. Hofmann, R. Wille, “A Fully Planar Approach to Field-coupled Nanocomputing: Scalable Placement and Routing Without Wire Crossings,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.
- [27] S. Hofmann, J. Drewniok, **M. Walter**, R. Wille, “MNT Designer: A Comprehensive Design Tool for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.
- [28] S. Hofmann, **M. Walter**, R. Wille, “Late Breaking Results: Physical Co-Design for Field-coupled Nanocomputing,” in *Design, Automation and Test in Europe (DATE)*, 2025.
- [29] S. Hofmann, **M. Walter**, R. Wille, “Physical Design for Field-coupled Nanocomputing with Discretionary Cost Objectives,” in *IEEE Latin American Symposium on Circuits and Systems (LASCAS)*, **Best Paper Award Candidate**, 2025.
- [30] S. S. H. Ng, **M. Walter**, J. Drewniok, S. Hofmann, R. Wille, K. Walus, “Building a Machine Learning Accelerator with Silicon Dangling Bonds: From Verilog to Quantum Dot Layout,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, **Best Student Paper Award**, 2025.
- [31] **M. Walter**, J. Drewniok, R. Wille, “Live Demonstration: An Application for Layout Resilience Analysis of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.
- [32] **M. Walter**, J. Drewniok, R. Wille, “The Operational Domain Explorer: A Comprehensive Framework to Unveil the Thermal Landscape of Silicon Dangling Bond Logic Beyond Conventional Operability,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.
- [33] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “On-the-fly Defect-Aware Design of Circuits based on Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. doi: 10.1109/NANO61778.2024.10628962
- [34] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Unifying Figures of Merit: A Versatile Cost Function for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. doi: 10.1109/NANO61778.2024.10628671
- [35] J. Drewniok, **M. Walter**, R. Wille, “The Need for Speed: Efficient Exact Simulation of Silicon Dangling Bond Logic,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. doi: 10.1109/ASP-DAC58780.2024.10473946
- [36] B. Hien, **M. Walter**, V. M. Santen, F. Klemme, S. S. Parihar, G. Pahwa, Y. S. Chauhan, H. Amrouch, R. Wille, “Technology Mapping for Cryogenic CMOS Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024. doi: 10.1109/ISVLSI61997.2024.00057
- [37] B. Hien, **M. Walter**, R. Wille, “Reducing Wire Crossings in Field-Coupled Nanotechnologies,” in *International Workshop on Logic Synthesis (IWLS)*, 2024.
- [38] B. Hien, **M. Walter**, R. Wille, “Reducing Wire Crossings in Field-Coupled Nanotechnologies,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. doi: 10.1109/NANO61778.2024.10628717
- [39] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, “Thinking Outside the Clock: Physical Design for Field-coupled Nanocomputing with Deep Reinforcement Learning,” in *International Symposium on Quality Electronic Design (ISQED)*, 2024. doi: 10.1109/ISQED60706.2024.10528711

- [40] S. Hofmann, **M. Walter**, R. Wille, “A* is Born: Efficient and Scalable Physical Design for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628808
- [41] S. Hofmann, **M. Walter**, R. Wille, “Late Breaking Results: Wiring Reduction for Field-coupled Nanotechnologies,” in *Design Automation Conference (DAC)*, 2024. DOI: 10.1145/3649329.3663491
- [42] S. Hofmann, **M. Walter**, R. Wille, “MNT Bench: Benchmarking Software and Layout Libraries for Field-coupled Nanocomputing,” in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546596
- [43] D. Marakkalage, **M. Walter**, S.-Y. Lee, R. Wille, G. De Micheli, “Technology Mapping for Beyond-CMOS Circuitry with Unconventional Cost Functions,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, **Best Presentation Award**, 2024. DOI: 10.1109/NANO61778.2024.10628909
- [44] S. S. H. Ng, J. Drewniok, **M. Walter**, J. Retallick, R. Wille, K. Walus, “Unlocking Flexible Silicon Dangling Bond Logic Designs on Alternative Silicon Orientations,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628802
- [45] **M. Walter**, J. Croshaw, S. S. H. Ng, K. Walus, R. Wolkow, R. Wille, “Towards Atomic Defect-Aware Physical Design of Silicon Dangling Bond Logic on the H-Si(100)-2x1 Surface,” in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546683
- [46] **M. Walter**, J. Drewniok, S. Hofmann, B. Hien, R. Wille, “The Munich Nanotech Toolkit (MNT),” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628747
- [47] **M. Walter**, J. Drewniok, R. Wille, “Ending the Tyranny of the Clock: SAT-based Clock Number Assignment for Field-coupled Nanotechnologies,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628908
- [48] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “QuickSim: Efficient and Accurate Physical Simulation of Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231266
- [49] J. Drewniok, **M. Walter**, R. Wille, “Minimal Design of SiDB Gates: An Optimal Basis for Circuits Based on Silicon Dangling Bonds,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633241
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